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## **EE/EX-5005(1)-CBGS**

### **B.E. V Semester**

Examination, June 2020

### **Choice Based Grading System (CBGS)**

### **Energy Conservation and Management**

*Time : Three Hours*

*Maximum Marks : 70*

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) Discuss the various energy problems. 7  
b) Briefly describe the instrument used for energy auditing. 7
2. a) What do you understand by “Energy Audit”? Write a note on “Indian energy scenario”. 7  
b) Illustrate the waste heat recovery techniques in an industry. 7
3. a) What is thermal insulation? Explain classification of thermal insulation. 7  
b) Explain the term DSM and how it can be achieved? Also discuss benefits of DSM. 7
4. a) What are functions of Lubricants? Explain the classification of lubricants. 7  
b) Explain time value of money. Also discuss payback period. 7

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5. a) An industrial consumer having a maximum demand of 200kW, maintains a load factor of 70%. The tariff rates are Rs. 800 per kVA of maximum demand per annum plus Rs. 2.20 per kWh of energy consumed. If the average power factor is 0.8 lagging, calculate the total energy consumed per annum and the annual electricity bill. 7  
b) What is motor efficiency? Explain the concept and importance of energy efficient motors. 7
6. a) What are the factors are to be considered while structuring a tariff plan for a utility? Discuss. 7  
b) Explain the causes of low power factor. Explain how power factor can be improved. 7
7. a) What is co-generation? Explain the benefits of co-generation. 7  
b) Describe the energy conservation process for a cement industry. 7
8. Explain the followings. (any two) 14
  - a) Energy efficient house keeping
  - b) Predictive and preventive maintenance
  - c) Energy conservation policy

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